Series: Method of Differences – Part 2

Questions 1–18

Question 1. Show that

$$\sum_{r=1}^{n} \frac{1}{r(r+1)} = \frac{n}{n+1}.$$

Question 2. Use the method of differences to find the value of $\sum_{r=1}^{40} \frac{1}{r(r+1)(r+2)}$, giving your answer in exact form.

Question 3. Use the method of differences to find the value of $\sum_{r=10}^{200} \frac{1}{r^2+5r+6}$, giving your answer in exact form.

Question 4. Use the method of differences to determine the value of $\sum_{r=5}^{55} \frac{1}{(r+1)(r+2)(r+3)}$, giving your answer in exact form.

Question 5. Show that

$$\sum_{r=10}^{n} \frac{1}{(r+3)(r+4)} = \frac{n-a}{b(n+c)},$$

where a, b and c are integers to be found.

Question 6. Use the method of differences to prove that for $n > 2$

$$\sum_{r=2}^{n} \frac{4}{4r^2-4r-3} = \frac{a(n-b)(an+b)}{cn^2-d},$$

where a, b, c and d are integers to be found.

Question 7.

a) Find an expression for $\sum_{r=1}^{n} \frac{2}{r^2+4r+3}$.

b) Hence determine the value of $\sum_{r=1}^{\infty} \frac{1}{r^2+4r+3}$.

Question 8. Find $\sum_{r=1}^{\infty} \frac{1}{(2r+1)(2r+3)}$.

Question 9.

a) Find an expression for $\sum_{r=1}^{n} \frac{3}{r^2+3r}$.

b) Hence determine the value of $\sum_{r=1}^{\infty} \frac{1}{r^2+3r}$.

Question 10. Find the smallest integer n such that $\sum_{r=3}^{n} \frac{1}{r^2-3r+2} > 0.45$.

Question 11.

a) Show that $\sum_{r=n}^{2n} \frac{1}{(3r-2)(3r+1)} = \frac{an+b}{(3n-2)(6n+1)}$, where a and b are integers to be found.

b) Hence find $\lim_{n\to\infty} \left(\sum_{r=n}^{2n} \frac{1}{(3r-2)(3r+1)} \right)$.

Question 12. A company is managing a reservoir with a total capacity of 5 million litres of water. To regulate water usage, the company models the amount of water released each year as follows:

$$W = \frac{15}{r^2+6r+8},$$

where W is the amount of water, in millions of litres, released in year r, where $r = 1$ represents year one.

a) Show that the total amount of water released in the first n years is given by $\frac{an^2+bn}{c(n+3)(n+4)}$, where a, b and c are constants to be found.

b) Explain why, according to this model, the reservoir will never run out of water.

c) The company decides to release an additional fixed amount of water each year to ensure that the entire 5 million litres is used up exactly after 30 years. Refine the formula for W to reflect this change.

Question 13. A government agency releases funding to startup businesses over time. The amount released in month r measured in millions of pounds (£) is given by

$$A_r = \frac{300}{(r+1)(r+4)}, \quad r = 1, 2, 3, \ldots,$$

where A_r is the amount of funding released in month r.

a) Show that the total amount of funding released in the first n months can be written in the form $A = \dfrac{25n(an^2 + bn + c)}{3(n+2)(n+3)(n+4)}$, where a, b and c are constants to be found.

b) Explain why, according to this model, the government agency expects not to release more than £109 million in total.

c) The agency claims that after 30 months, at least 90% of the total funding will have been released. Verify this claim.

Question 14. Let $C_r = \dfrac{\sin 1}{\cos r \cos(r+1)}$, where r is a positive integer.

a) By considering $\sin r \cos(r+1) - \cos r \sin(r+1)$, show that $C_r = \tan(r+1) - \tan r$.

b) Hence evaluate $\displaystyle\sum_{k=1}^{n} C_r$.

Question 15. Let $D_r = \dfrac{\sinh 1}{\sinh(r)\sinh(r+1)}$, where r is a positive integer.

a) By considering $\sinh(r+1)\cosh r - \cosh(r+1)\sinh r$, show that $D_r = \coth r - \coth(r+1)$.

b) Hence evaluate $\displaystyle\sum_{k=1}^{n} D_r$.

Question 16.

a) Use $\tan(A-B) = \dfrac{\tan A - \tan B}{1+\tan A \tan B}$ with $A = (r+1)x$ and $B = rx$ to show that

$$\tan(rx)\tan((r+1)x) = \frac{\tan((r+1)x)}{\tan x} - \frac{\tan(rx)}{\tan x} - 1.$$

b) Hence show that

$$\tan\frac{2\pi}{20}\tan\frac{3\pi}{20} + \tan\frac{3\pi}{20}\tan\frac{4\pi}{20} + \cdots + \tan\frac{14\pi}{20}\tan\frac{15\pi}{20} = \frac{\tan\frac{15\pi}{20}}{\tan\frac{\pi}{20}} - 15.$$

Question 17. A sequence u_r is defined by $u_r = \ln(r+1)$, where $r \geq 1$.

a) Show that $u_{r+1} - u_r = \ln(1 + \frac{1}{r+1})$.

b) Show that $u_{r+1} - u_r = \displaystyle\int_{r+1}^{r+2} \frac{1}{x}\, dx$ and hence, using the Mean Value Theorem for integrals, show that $\dfrac{1}{r+2} < u_{r+1} - u_r < \dfrac{1}{r+1}$.

c) Using method of differences, show that $\displaystyle\sum_{r=1}^{n}(u_{r+1} - u_r) = \ln(n+2) - \ln 2$.

Question 18. A sequence is defined by $u_r = y(r)$ where $y(x)$ satisfies the differential equation $\dfrac{dy}{dx} = \dfrac{1}{x(x+1)}$ for $x > 0$.

a) Find the general solution to the differential equation.

b) Hence find an expression for $u_{r+1} - u_r$.

c) Use the method of differences to evaluate $\displaystyle\sum_{r=1}^{n}(u_{r+1} - u_r)$, giving your answer as a single term.

d) Hence, find $\displaystyle\sum_{r=1}^{\infty}(u_{r+1} - u_r)$.